

# Zihan Liang

Undergraduate AI Researcher at Xi'an Jiaotong-Liverpool University

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## Research Profile

Undergraduate researcher in artificial intelligence developing reproducible machine-learning methods for mixed-frequency time-series forecasting and sequential social-media risk assessment. Experience spans Transformer and attention architectures, NLP sequence modeling, multi-horizon and imbalanced-class evaluation, interpretability, and research communication through an SSRN preprint and an ICSC poster.

## Education

**Xi'an Jiaotong-Liverpool University (XJTLU)**

Sep 2024 – Jul 2028 (expected)

BEng Artificial Intelligence (Intelligent Systems)

Suzhou, China

Dual Degree: BSc (Hons), University of Liverpool (expected)

Liverpool, UK

– **Academic standing:** Completed Stage 2; Stage 2 weighted average 78/100 and overall weighted average 74/100 (First-Class range under XJTLU's British marking criteria).

– **Selected marks:** Python for AI 92, Maths for Machine Learning 86, Engineering Mathematics I 82, AI Fundamentals & Ethics 81, and Data Structures and Algorithms 78.

## Research Outputs

- Q. Luo, R. Xu, **Z. Liang**, Z. Hou, and R. Zhao. *Do Macroeconomic and Geopolitical Risk Factors Improve FX Volatility Forecasts? A Mixed-Frequency Transformer Approach*. SSRN preprint, 2025. [Preprint](#) | [DOI](#).
- **Z. Liang**, R. Xu, and R. Zhao. *Public Mental Health Monitoring: Detecting Suicide Risk in Sequential Social Media*. Conference poster presented at ICSC 2025. [Code](#) | [Poster](#).

## Research Experience

**Mixed-Frequency Transformer for USD/CNH Volatility Forecasting**

Aug 2025

Model Developer, XJTLU

Suzhou, China

- Implemented temporal and positional encodings and contributed to the Transformer architecture for multi-horizon FX-volatility forecasting.
- Proposed cross-attention using daily technical representations as queries and macroeconomic/geopolitical streams as keys and values; evaluated 1-, 5-, and 10-day horizons.
- Produced pooled and horizon-specific evaluations, predicted-versus-actual plots, model-pipeline figures, and Integrated Gradients analyses of the 2024 U.S. election period.

**Sequential Suicide-Risk Prediction from Social Media**

Nov 2025

Lead Student Researcher; First Author of ICSC Poster, XJTLU

Suzhou, China

- Built the RSD-15K pipeline from data acquisition and exploratory analysis through preprocessing, model training, and user-level evaluation.
- Developed a hierarchical architecture combining Twitter-RoBERTa, Bi-GRU, and attention; applied emoji/token normalization, head-tail retention, and weighted focal loss.
- Achieved weighted F1 0.46 versus 0.43 for a RoBERTa baseline under five-fold user-level evaluation; completed an IEEE BigData Challenge submission and presented an ICSC poster.

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|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>The Fairness Paradox: A Stratified Hybrid-Merit Model</b>                                                                                                                 | Jan 2026            |
| Team Member, Mathematical Contest in Modeling                                                                                                                                | COMAP               |
| – Co-authored a team framework combining inverse inference, counterfactual scoring-rule analysis, Kruskal–Wallis testing, and mechanism design for fair competition scoring. |                     |
| – Team designated as a 2026 MCM Finalist.                                                                                                                                    |                     |
| <b>Grouped NIPT Timing Optimization and Abnormality Detection</b>                                                                                                            | Sep 2025            |
| Statistical Modeling Contributor, CUMCM 2025                                                                                                                                 | Jiangsu, China      |
| – Developed GLM/GAM analysis and a grouped testing-time recommendation strategy using BMI and gestational-age patterns.                                                      |                     |
| – Built logistic classification with ROC/Youden thresholding for abnormality detection; team received Jiangsu Provincial Second Prize.                                       |                     |
| <b>Au<sub>20</sub> Scientific Visualization Support</b>                                                                                                                      | Oct 2025            |
| Visualization and Reporting Contributor, XJTU                                                                                                                                | Suzhou, China       |
| – Built Python utilities for atomic-cluster structures and bond diagrams and produced animated LaTeX/PDF presentation assets. <a href="#">Code</a> .                         |                     |
| <b>Home Repair Assistant Robotics</b>                                                                                                                                        | Jun 2025 – Aug 2025 |
| Engineering Contributor, XJTU SURF                                                                                                                                           | Suzhou, China       |
| – Configured the ROS 2 runtime, verified node-level execution, and deployed and validated the codebase on NVIDIA Jetson Orin Nano.                                           |                     |

## Selected Honors and Awards

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- **Finalist**, Mathematical Contest in Modeling, 2026.
- **Jiangsu Provincial Second Prize**, Contemporary Undergraduate Mathematical Contest in Modeling, 2025.
- **Selected RLLS Honors**, 2025: Grand Prize (Au<sub>20</sub> delta-learning), First Prize (NIPT modeling), and Second Prizes (FX forecasting; suicide-risk modeling).
- **Jiangsu Provincial Third Prize**, “Keyou Cup” University Intelligent Robot Creativity Competition, Home Repair Assistant, 2025.

## Selected Presentations

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- **ICSC 2025, Poster Presenter:** Presented *Public Mental Health Monitoring: Detecting Suicide Risk in Sequential Social Media*.
- **XJTU RLLS 2025, Poster and Oral Presenter:** Delivered a 10-minute presentation and 5-minute Q&A on hierarchical suicide-risk modeling; received Second Prize.

## Technical Skills

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**Programming and tools:** Python (PyTorch, Hugging Face Transformers, scikit-learn, NumPy, Pandas, Matplotlib), C/C++, Git, LaTeX.

**Methods:** Transformers, attention mechanisms, mixed-frequency and multi-horizon forecasting, NLP sequence modeling, cross-validation, imbalanced classification, statistical modeling, and scientific visualization.

**Languages:** Chinese (native); English (English-medium degree programme; official EMI letter available).